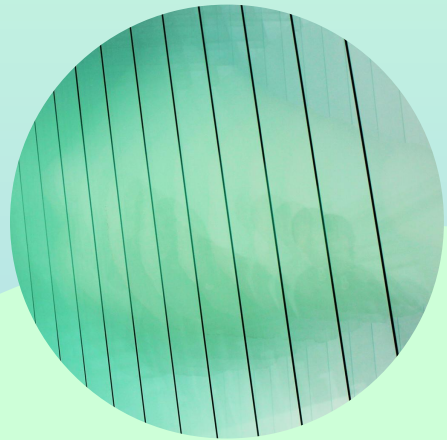


Theoretical Framework for machine Learning Researcher



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ML with Me



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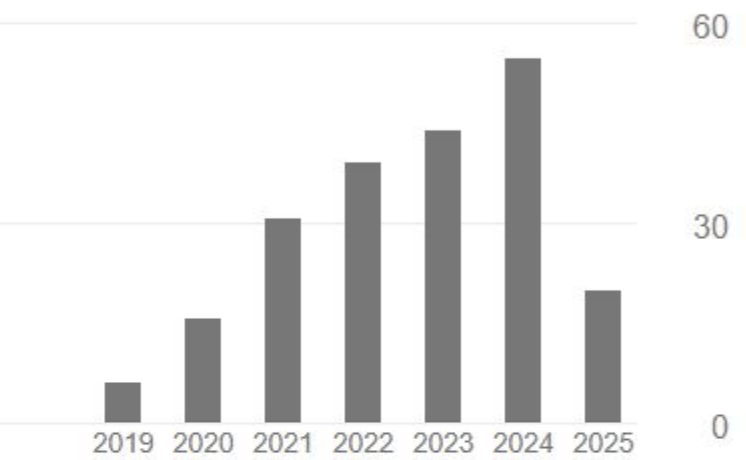
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☐	The design of SDN based detection for distributed denial of service (DDoS) attack MM Oo, S Kamolphiwong, T Kamolphiwong 2017 21st International Computer Science and Engineering Conference (ICSEC), 1-5	36	2017
☐	Analysis of features dataset for DDoS detection by using ASVM method on software defined networking MM Oo, S Kamolphiwong, T Kamolphiwong, S Vasupongayya International Journal of Networked and Distributed Computing 8 (2), 86-93	10	2020
☐	SDN based Design for Detection of Distributed Denial of Service (DDoS) attack (インターネットアーキテクチャ) S Kamolphiwong, T Kamolphiwong 電子情報通信学会技術研究報告= IEICE technical report: 信学技報 117 (299), 13-18		2017
☐	SDN based Design for Detection of Distributed Denial of Service (DDoS) attack MM Oo, S Kamolphiwong, T Kamolphiwong IEICE Technical Report; IEICE Tech. Rep. 117 (299), 13-18		2017

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What is Theoretical Framework?

A theoretical framework is a *foundational review of existing theories* that serves as a roadmap for developing the arguments you will use in your own work.

Why

How



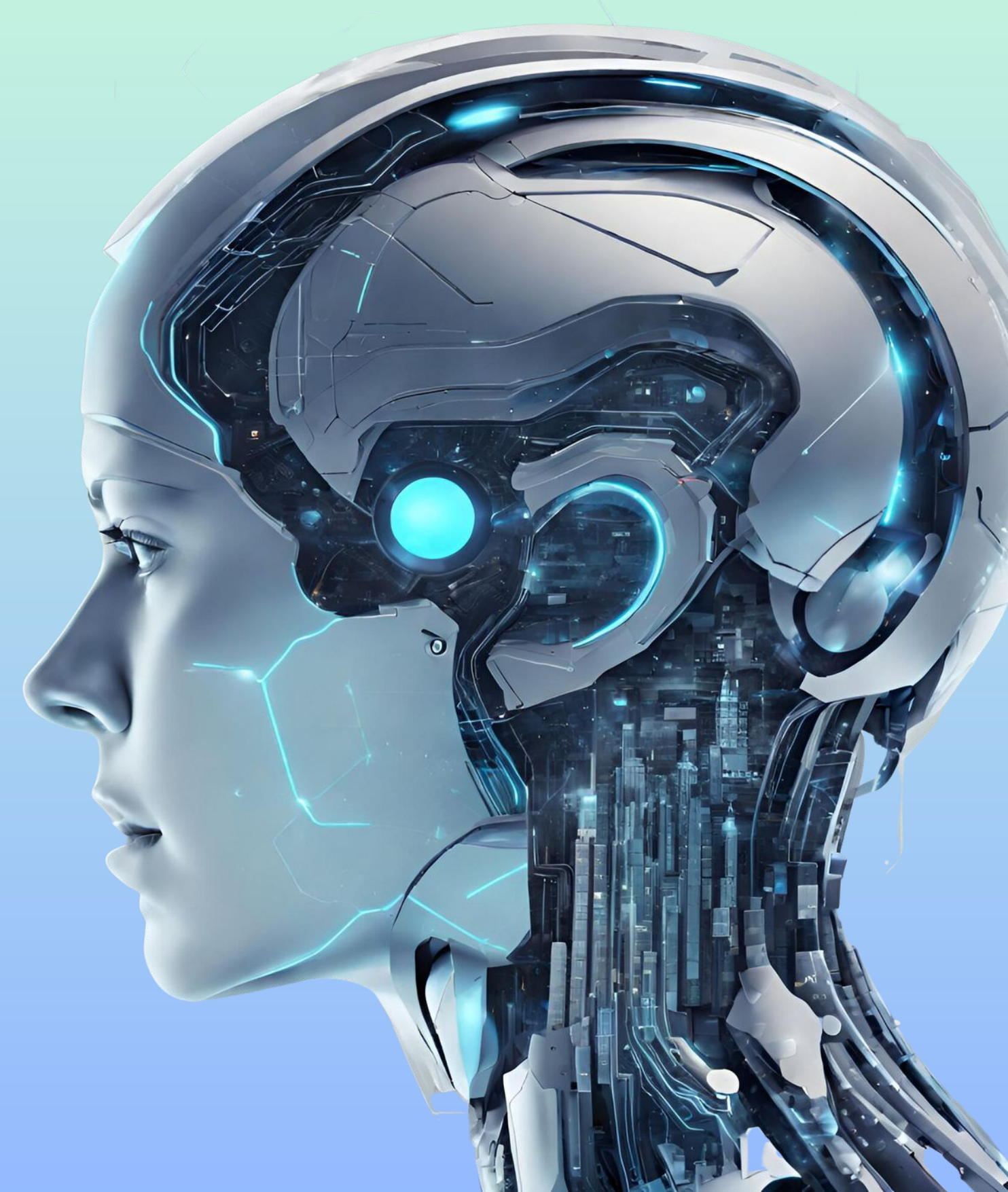
Theoretical Framework for ML

- Statistical Learning Theory
- Optimization Theory
- Information Theory
- Algorithmic Computational Complexity



Statistical Learning Theory

- Framing the problem
- Data Understanding
- Data Cleaning
- Data selection and preparation
- Model evaluation
- Configuration of the model
- Model selection and presentation
- Model predictions



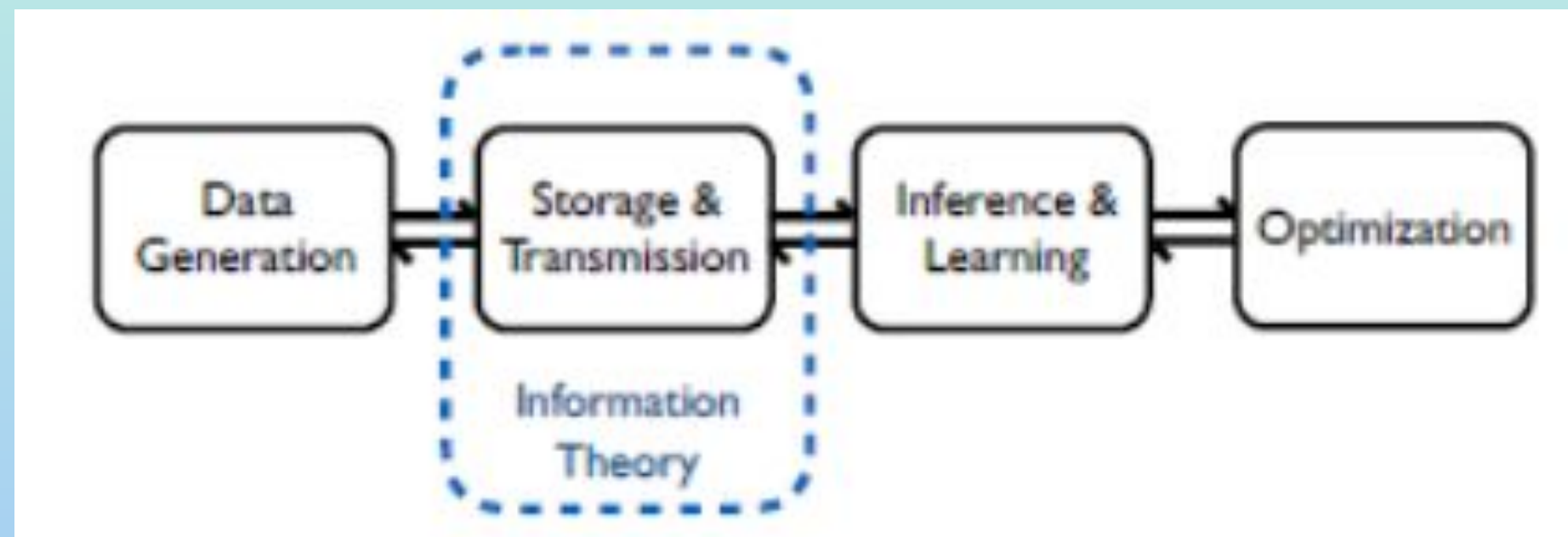
Optimization Theory in Machine Learning

- Optimization Theory in machine learning are mathematical techniques used to adjust a model's parameters to minimize errors and improve accuracy.
- This can help models learn from data by finding the best possible solution through iterative updates.

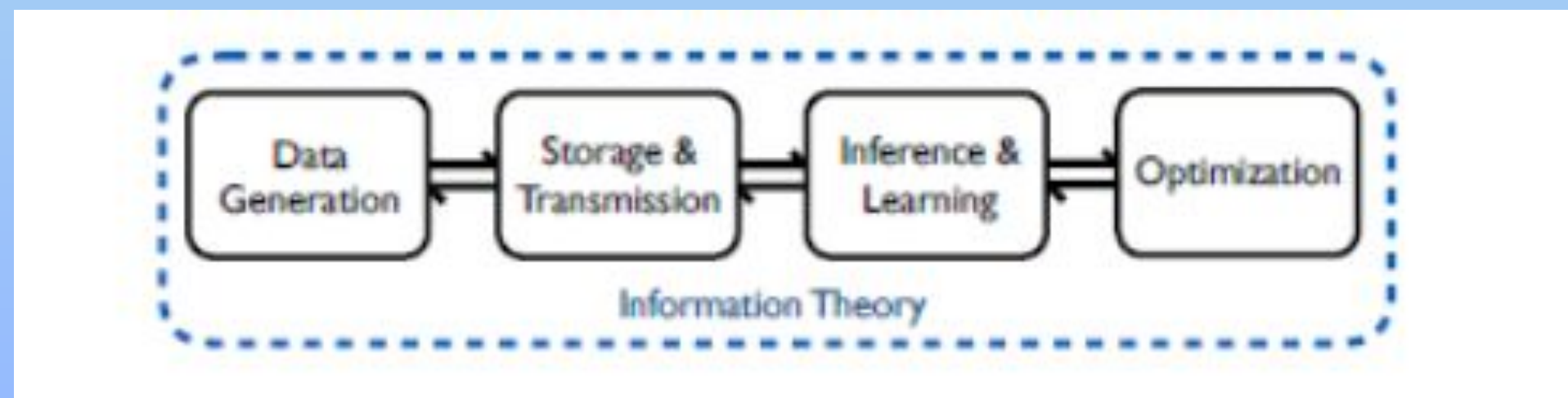


Information theory and Data

Information theory is a theory of communication



Information theory is a theory of communication



Key Concepts of Information Theory

- **Entropy** measures the uncertainty or unpredictability of a random variable.

In machine learning, entropy quantifies the amount of information required to describe a dataset.

- **Mutual information** measures the amount of information obtained about one random variable through another random variable. It quantifies the dependency between variables.

Algorithmic Computational Complexity

- Algorithmic computational complexity is the measure of the resources, typically time and space, an algorithm requires relative to its input size.
- It helps analyze and compare the efficiency of algorithms, focusing on how their performance scales with increasing input.
- This is often expressed using Big-O notation to describe the upper bound or "worst-case" performance, as seen in functions like $O(n)$ (linear) or $O(n^2)$ (quadratic)

Summary

- “Theory is what turns data into knowledge”

⋮

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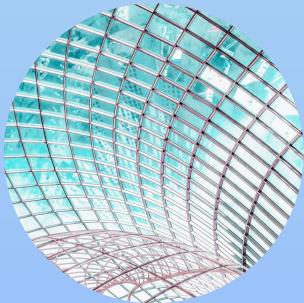
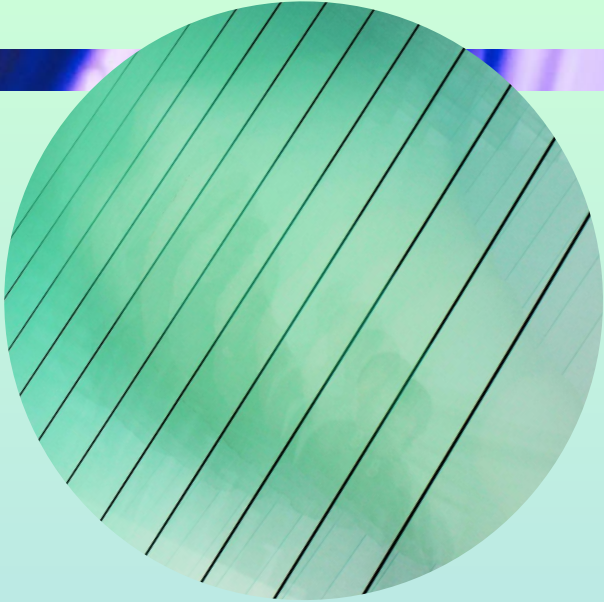
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Thank you!